

Three Models, One Lesson: What Happens When You Point a Geospatial Foundation Model at Philadelphia's Rooftops

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The Setup

Philadelphia Licenses and Inspections maintains a registry of buildings that are structurally compromised. Roughly 140 parcels at any given time carry the most severe designation - "Imminently Dangerous" - designating the structure as at imminent risk of collapse, with another 7,800 flagged as "Unsafe." Together they account for less than 1.5 percent of the city's roughly 547,000 parcels, represented in Figure 1 as a handful of red and orange points scattered across a dense stable background.

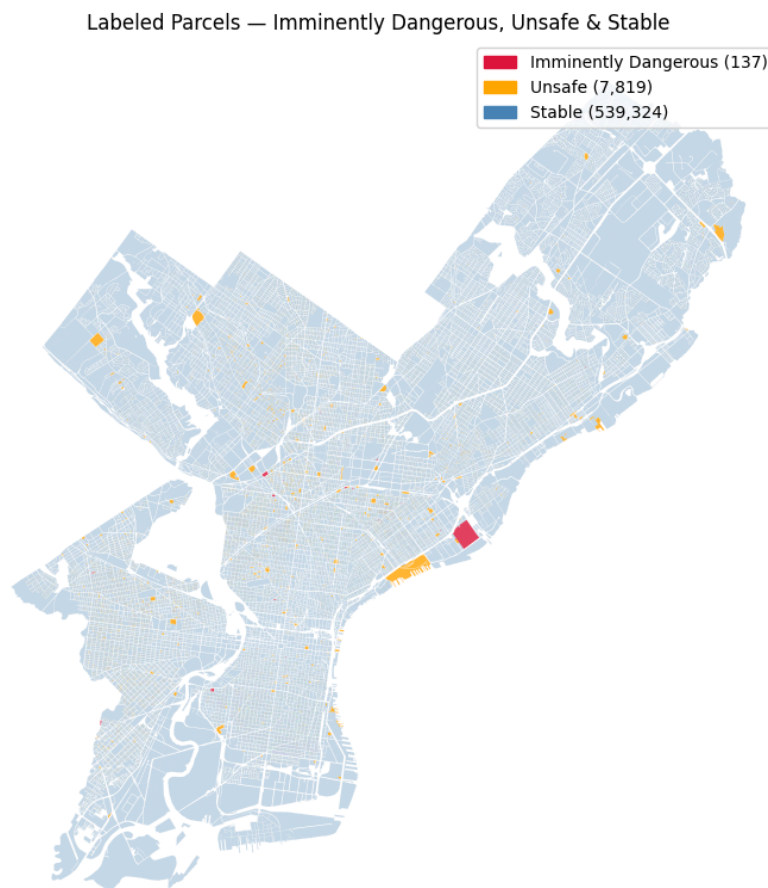


Figure 1: Spatial distribution of Imminently Dangerous (red), Unsafe (orange), and Stable (blue) properties across Philadelphia.

This building licensing inspections registry is reactive, responding to complaints, inspection triggers, and permit activity rather than any systematic, visual review of the built environment. Meanwhile, the city publishes annual 0.25 ft (7.62 cm) aerial imagery encompassing every parcel in the city, providing an unprecedentedly detailed view of the city from above. Remote sensing has historically been utilized in scenarios such as classifying natural disaster severity and identifying roads and built structures, but could this be taken a step further at a much finer level of detail?

We attempted to answer the question of whether Clay's Foundation Model v1.5 could be utilized in concert with this aerial imagery to look at rooftops across Philadelphia and determine whether a deep learning approach could be viable for preliminary building inspection. This post walks through three attempted architectures, and while none of them produced a deployable classifier, together they say something concrete about the transfer limits of Earth observation foundation models when pushed outside their pretraining distribution.

Evaluation Setup: Protocol established, held constant across iterations

Before we conducted any modeling, we fixed our data split so the three approaches would be directly comparable. Ten percent of all parcels were set aside as a natural-distribution holdout, maintaining an approximately 1.4 percent distressed-to-stable parcel ratio. The remaining 90 percent became the training pool, rebalanced to a 2:1 stable-to-distressed ratio through data augmentation and then split 80/20 stratified into train and validation folds. Every model sees exactly the same chips, the same labels, and the same geography. Balanced validation F1 is useful for threshold tuning, but the numbers that actually matter are always the ones on the natural-prior holdout, because that is the only split that reflects operational reality.

Chips are 130-foot windows around each parcel centroid, padded to a square resolution and resized to whatever input dimensions the backbone expects. Labels collapse L&I's three-class coding into binary: zero for stable, one for any property the city has flagged as distressed.

Version 1: Clay v1.5 with a frozen backbone and an MLP head

The first approach was the one Clay recommends for downstream tasks on a foundation model. We loaded Clay's v1.5 MAE encoder, consisting of over 312 million parameters with Vision Transformer-Large, froze it, and attached a small MLP head ($1024 \rightarrow 512 \rightarrow 256 \rightarrow 1$) trained with BCE loss. Inputs were normalized with Clay's LINZ band statistics because LINZ, at 0.5 m, is the closest pretraining platform to Philadelphia's 0.3 m aerial. Optimization was AdamW at $5e-5$ with a OneCycle

warmup in bf16, and the augmentation stack included flips, 180-degree rotation, resized crops, and color jitter.

Training loss plateaued at approximately 0.636 within three epochs and refused to move. That number is suspicious on its own — it is essentially the entropy of the class prior — so we ran a diagnostic. For each batch, we measured `std_across_batch` on the 1024-dimensional CLS output: the dimension-wise standard deviation of the embeddings across visually different chips. The value came back at roughly 0.03. Clay was producing near-constant embeddings regardless of input. On the natural holdout, recall oscillated between 0.001 and 0.333 depending on where the threshold happened to fall on pure noise.

The root cause was not the head. Clay v1.5 was pretrained on imagery ranging from 0.5 m LINZ aerial down through 10 m Sentinel-2. Philadelphia's 0.3 m aerial is outside every platform in Clay's metadata. At that resolution, what fills a chip is no longer urban morphology in the satellite sense — it is individual shingles, gutter lines, and asphalt grain. The foundation model's filters have never been asked to care about those features, and they do not respond to them.

Version 2: ConvNeXt-Tiny with layer-wise learning rate decay

The natural pivot from this failure was to abandon the satellite foundation model and try a general-purpose vision backbone. ConvNeXt-Tiny, 28 million parameters, pretrained on ImageNet-22k and fine-tuned to ImageNet-1k, is a standard choice for this kind of drop-in. We unfroze the full network, attached a lightweight head (LayerNorm → Linear(768→256) → Linear(256→1)), and used layer-wise learning rate decay with a decay factor of 0.5 per stage — head at $5e-4$, stem at roughly $6.25e-5$. Everything else matched v2: same chips, same augmentations, same loss, same splits.

This was the first version where training loss moved past the prior entropy. Validation AUC became non-trivial, and the PR-curve-tuned operating point transferred to the natural-prior holdout rather than collapsing on contact with the real class distribution. The result was not extraordinary — the model is still working against a label column that is only partially visible from overhead — but it was coherent.

The methodological point is worth stating plainly: for sub-meter urban RGB, ImageNet pretraining outperformed the satellite-foundation-model pretraining on the same pixels. At 0.3 m, a row home rooftop is textured, material-heavy, and geometrically regular in ways that resemble natural-image content more than satellite content. ImageNet's filters have learned those textures from 1.3 million photographs; Clay has not.

Version 3: back to Clay, this time by the book

Before we declared Clay unsuitable for the task, we wanted to rule out a specific alternative explanation. Clay's official documentation does not prescribe neural fine-tuning for classification. Its [finetune-on-embeddings tutorial](https://clay-foundation.github.io/model/finetune/finetune-on-embeddings.html) runs Clay as a pure feature extractor, caches CLS tokens per chip, and trains a scikit-learn RandomForest on those cached vectors. The tutorial reaches 90 percent accuracy on a marina-detection task with 216 samples. If v2 had failed because we deviated from Clay's documented usage pattern, v4 would catch it.

We followed the recipe exactly. ``ClayMAEModule.load_from_checkpoint`` with the prescribed ``model_size="large"`, `dolls=[16,32,64,128,256,768,1024]`, LINZ conditioning, zero time and latitude/longitude vectors as the docs sanction. The classifier was a 500-tree RandomForest with `class_weight="balanced"`. We went further than the tutorial in places, using a 3072-dimensional readout that concatenated CLS with mean-pooled and max-pooled patch tokens, and averaging over five rotation-and-flip test-time augmentations. We also compared against a linear probe and an isotonic-calibrated RF to rule out the classifier family as a confound.`

At the default 0.5 probability threshold, the classifier predicts zero distressed parcels in the balanced validation set.

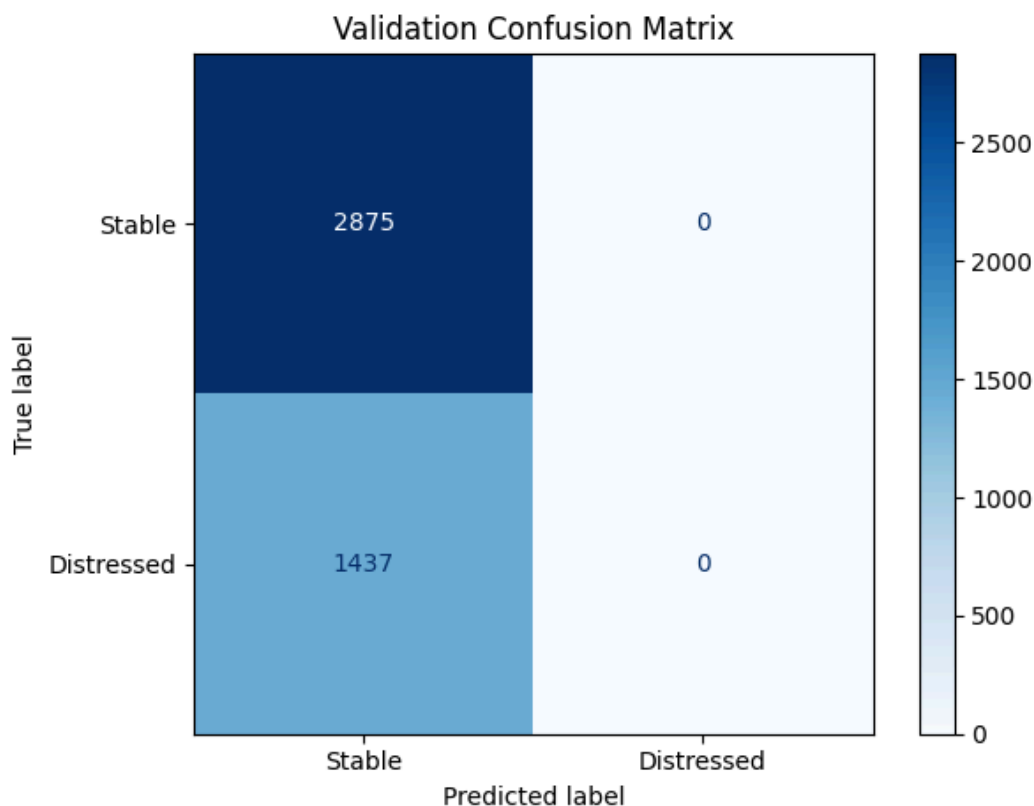


Figure 2: Validation confusion matrix at threshold 0.5 — all 1,437 distressed val parcels are classified as stable

At the F1-tuned threshold of 0.23, applied to the natural-distribution holdout, it predicts distressed on every parcel — 54,728 positives, zero negatives, recall 1.0, precision 0.014.

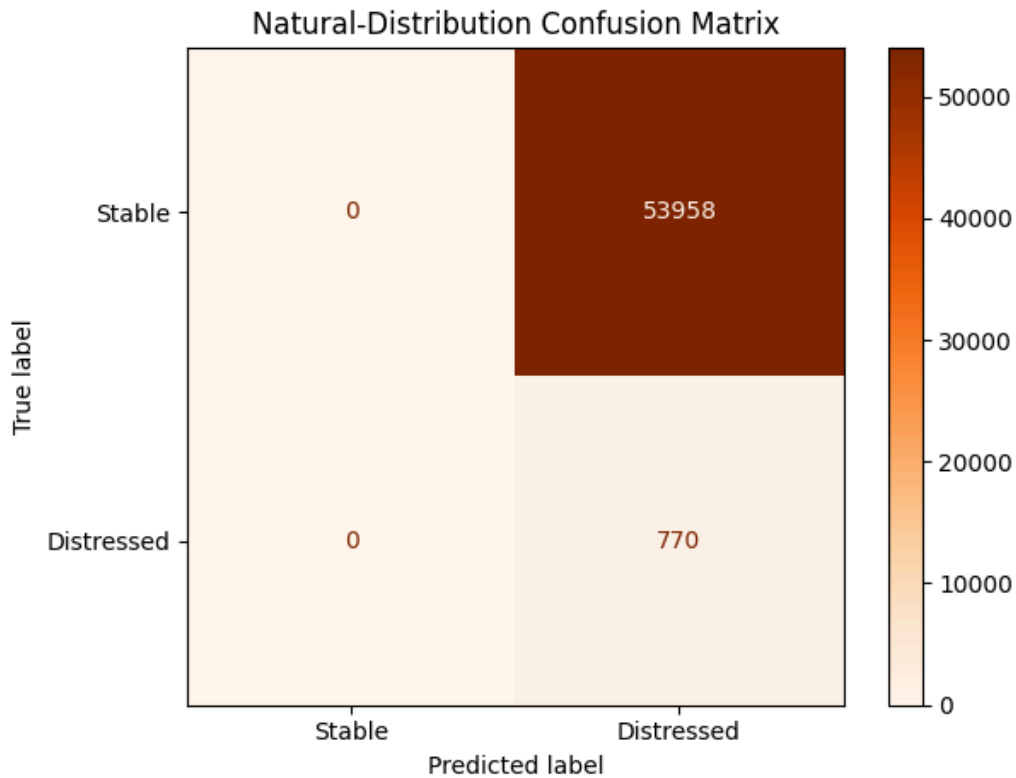


Figure 3: Natural-distribution confusion matrix at tuned threshold 0.23 — the classifier flips to flagging every parcel in the city.

There is no threshold between those two collapses that yields a usable classifier. This is not a tuning problem, and it is not a classifier-capacity problem. It is a statement about the features themselves.

What the embeddings actually look like

Two diagnostics from the v4 run make the underlying geometry visible. A t-SNE projection of the validation CLS embeddings is the first:

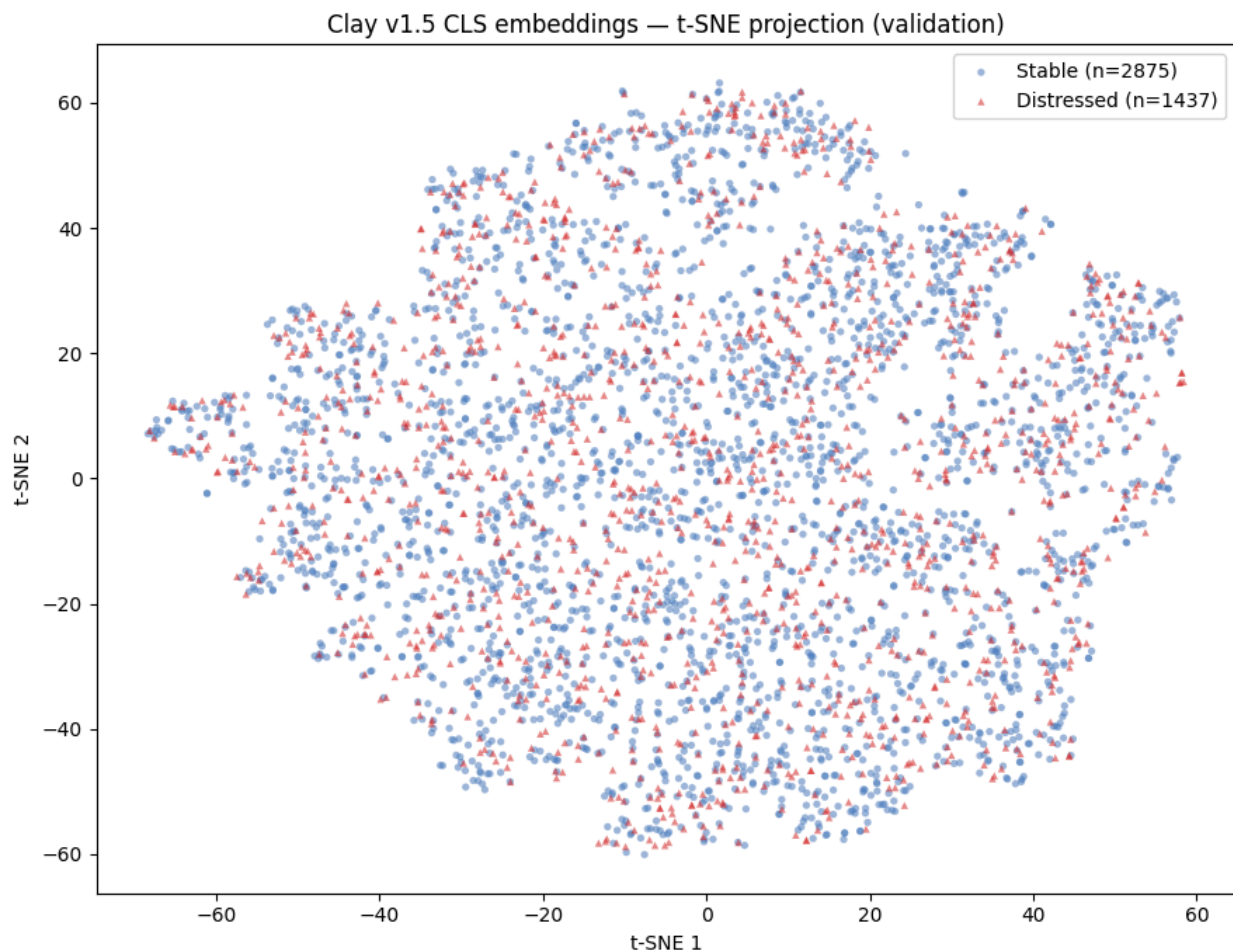


Figure 4: t-SNE of Clay v1.5 CLS embeddings on the validation set — stable and distressed points are uniformly intermixed.

Stable (blue) and distressed (red) are uniformly intermixed. There is no region of Clay's 1024-dimensional embedding space that concentrates distressed parcels. No classifier operating on these features — a Random Forest, a linear probe, a neural head, any of them — can separate the classes, because the separating geometry is not there to be learned.

The Random Forest's feature importances give the same story in a different register. In a feature space that carries real signal for a task, a tree ensemble typically concentrates its splits on a small number of informative dimensions, producing a sharp power-law falloff in the sorted importance distribution. What we see instead is a nearly flat spread across all 1,024 dimensions, ranging from about 0.0013 at the top to 0.0006 at the bottom.

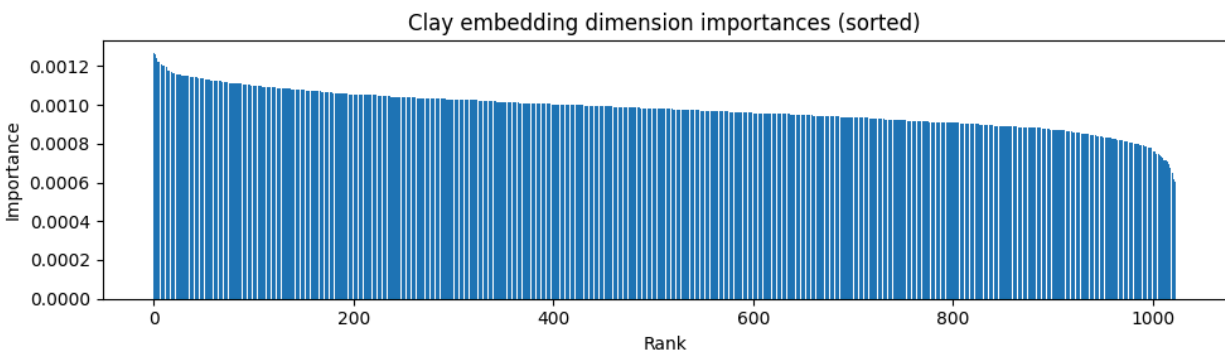


Figure 5: Sorted Clay embedding dimension importances from the RandomForest classifier — nearly uniform across 1,024 dimensions, with no informative subset standing out.

This is the classifier telling us it cannot find any useful dimensions, so it is averaging across all of them and hoping for the best. Combined with the t-SNE, the picture is consistent: the features do not contain a separating signal, and any classifier trained on them is effectively fitting noise.

A third diagnostic is cosine nearest-neighbor retrieval. Given a known-distressed query chip, we pull the six closest chips in Clay's embedding space:

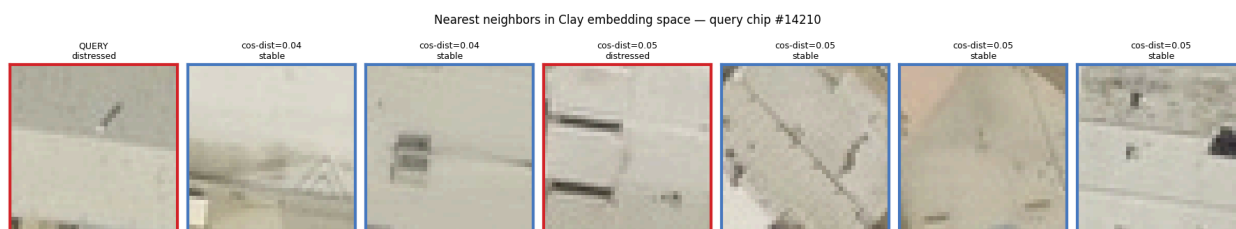


Figure 6: Nearest neighbors in Clay embedding space — five of six neighbors at cosine distances of 0.04 are stable, despite strong visual similarity to the distressed query.

Five of the six neighbors are stable, at cosine distances around 0.04. All are pale flat roofs with similar orientation and material, showing that Clay's features capture roof material and geometry rather than capturing conditions. The distress label and the visual-similarity label are not the same thing in this feature space, which means they cannot be the same thing at the output of any classifier built on top.

A final view, the full natural-distribution holdout colored by predicted probability, shows the same story at city scale:

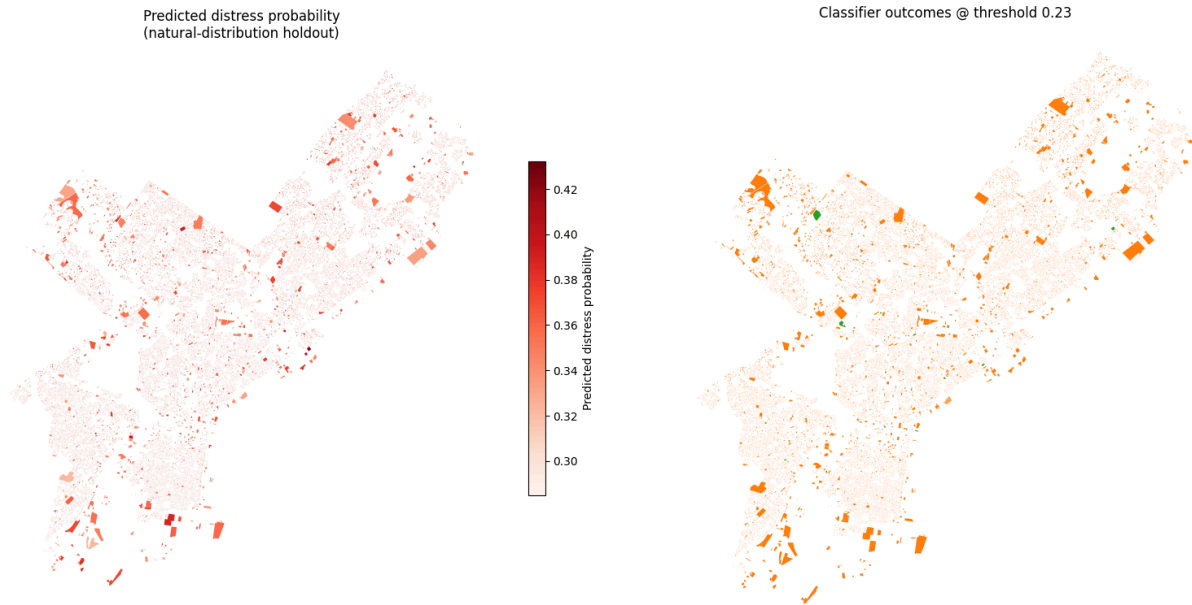


Figure 7: Predicted distress probability across Philadelphia's natural-distribution holdout, with classifier outcomes at threshold 0.23.

Predicted probabilities cluster in a narrow band between roughly 0.28 and 0.43 across the entire city. There is no high-confidence tail on either end, which is precisely why any threshold either catches everyone or catches no one.

Takeaways

For 0.3 meter, single-date, aerial RGB imagery, this approach is not sufficient even with adapting a supposedly sensor-agnostic foundation model. Clay's model was developed utilizing sensors with resolutions from 0.5 m LINZ through 10 m Sentinel-2, which largely pre-disposes the model to detecting urban morphology. At a finer resolution, the content of our training chip included individual shingles and gutter lines, not urban morphology, and Clay's pretraining distribution never sees imagery where those are the dominant features. Three different classifier heads on Clay features (MLP with a frozen backbone, linear probe, calibrated Random Forest) all fail in structurally similar ways on the same data, while a general-purpose ImageNet backbone fine-tuned on the same chips produces a working classifier. Pretraining distance, not head architecture and not classifier capacity, is the bottleneck.

Furthermore, the distress label is partially unobservable from directly above. L&I violations include foundation cracks, interior water damage, failed egress, and facade conditions that never rise above the roofline. Our third model iteration, utilizing nearest-neighbor analysis, puts a ceiling on how much any purely overhead-RGB classifier can achieve. Clay reliably retrieves visually similar parcels, and those

parcels carry mixed labels, because the label is not fully determined by what the imagery shows. This cap applies to our first iteration as well.

Does temporal differencing rescue the task?

One possible next move could be to add a temporal signal to pull out visually distinct changes in building quality over time. A building that collapsed between 2024 and 2025 would look different with a new tarp, fresh debris, or a hole where a roof used to be. We computed four chip-level change features between the 2024 and 2025 aerial mosaics: mean absolute difference and 95th-percentile absolute difference over pixel values, structural similarity (SSIM) between the two dates, and `dark\_new\_frac`, the fraction of pixels that appeared newly dark in 2025 (a rough proxy for holes and shadows from collapsed structures).

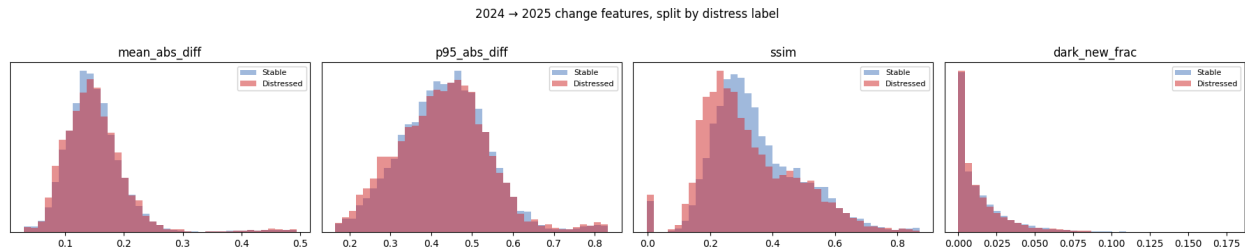


Figure 8: 2024 to 2025 change feature distributions for stable (blue) and distressed (red) parcels. All four features show nearly complete overlap between the two classes.

Stable and distressed parcels are nearly indistinguishable on all four features (Figure 8). There is a small shift in SSIM (distressed parcels have slightly lower structural similarity between years, as we might hope) and a small tail in `dark\_new\_frac` that leans distressed, but nothing that approaches separability.

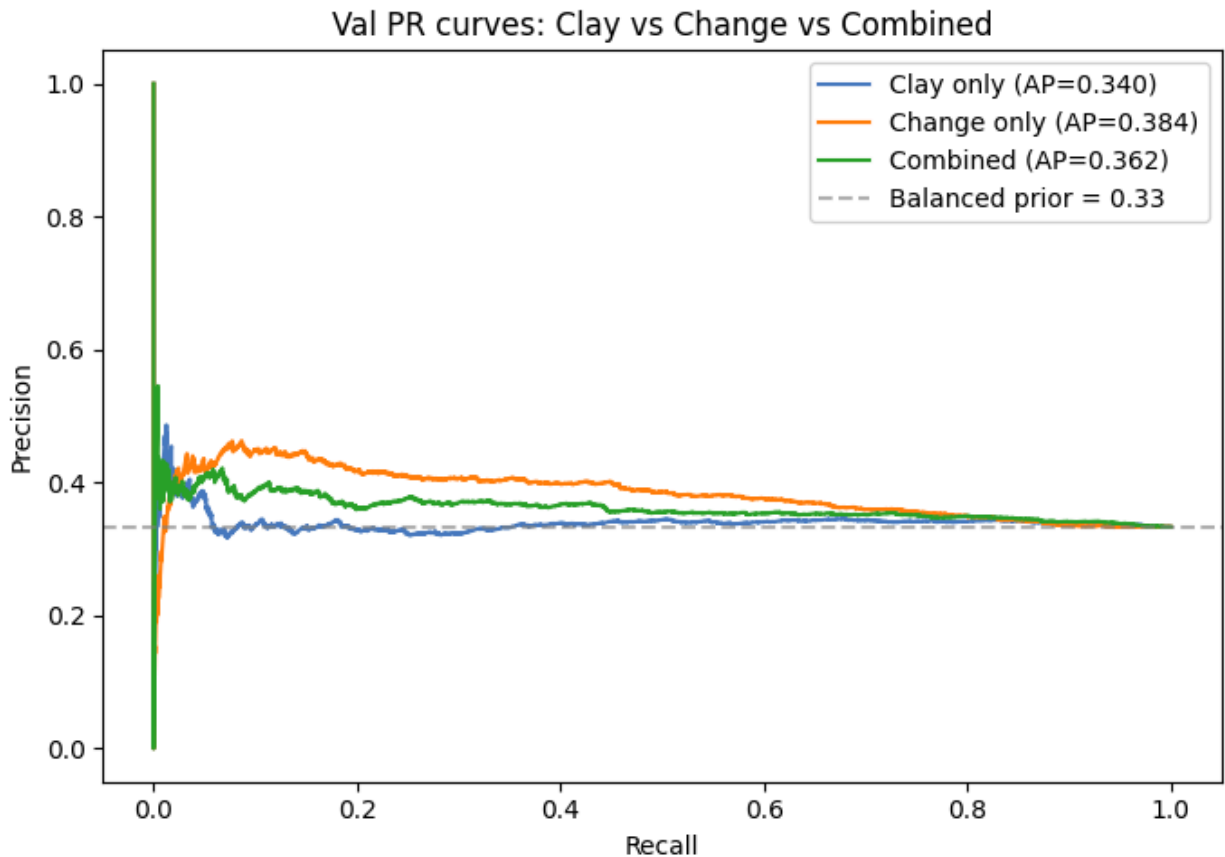


Figure 9: Validation precision-recall curves for Clay features only, change features only, and combined. All three hover just above the balanced-prior baseline of 0.33.

The cleanest test of these features is to put them in a precision-recall curve against the Clay embeddings alone and against a combined Clay + change model (Figure 9). All three AP values sit within a few points of the balanced prior of 0.33 — Clay alone at 0.340, Change alone at 0.384, Combined at 0.362. The change features edge out Clay on average precision, which tells us they do contain marginally more signal than the frozen embedding, but none of the three approaches carries enough information to lift precision meaningfully above the prior at any useful recall.

This is a harder negative result than the single-date one, because it closes off the most obvious complementary signal. Temporal differencing on 0.3 m RGB is not the missing piece. The change features capture real pixel-level variation between years, but that variation is dominated by lighting, shadow angle, tree canopy seasonal effects, and vehicle placement — not by structural condition changes at a rate that differs between stable and distressed parcels.

Next Steps

Moving forward, with temporal differencing on RGB now in the evidence pile as a third failed approach, the remaining extensions have to reach further. Non-imagery features that the L&I label genuinely depends on, such as parcel age, last inspection date, permit history, and adjacency to previously distressed parcels would let a classifier exploit the reactive structure of the label set instead of fighting it.

We would also treat our struggles with the Clay foundation model as an indication of the need for alternative deep learning models for this use case. The transfer limits of Earth-observation foundation models at sub-meter scales are not well characterized in the current literature, and further testing would require computation resources that exceed what we currently have available to us. While our results from this project were not the one we expected, but with additional time and resources we believe there is a path forward.

Code, embeddings, and the full evaluation notebooks are available in the project repository. Thanks to Nissim Lebovits and Guray Erus for the feedback throughout the semester.